

Resilient Manufacturing System

Redesign a production process so critical orders continue after one machine or supplier failure.

Customer Orion Aerospace Systems	Department Business Continuity	Difficulty Advanced	Time 2-3 class periods
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Production Goal: Continue priority production after a critical failure.

Background

A single machine currently performs a critical operation for an aerospace component. A recent outage stopped the entire line. Your team must create a resilience plan using alternate routing, inventory, or recovery strategies.

Learning Objectives

- Map dependencies in a production system
- Analyze single points of failure
- Develop contingency and recovery strategies
- Evaluate resilience tradeoffs

Equipment

- Current process map
- Machine and supplier capacity data
- Failure scenario cards
- Risk matrix
- Spreadsheet or planning board

Student Instructions

- 1 Create a current-state process map from raw material to shipment.
- 2 Mark every machine, supplier, data system, and inventory dependency.
- 3 Identify at least three single points of failure.
- 4 Draw one failure scenario and calculate lost production if no action is taken.
- 5 Develop three resilience options, such as alternate machine, safety stock, cross-training, or backup supplier.
- 6 Estimate cost, recovery time, and production impact for each option.
- 7 Use a weighted decision matrix to select a strategy.
- 8 Create a response playbook with detection, containment, alternate operation, and recovery steps.
- 9 Test the playbook using a second surprise failure scenario.

Engineering Requirements

- Address at least three failure points
- Maintain priority production in the selected scenario
- Include detection and communication steps
- Quantify recovery time and production impact

Constraints & Safety

- Use provided capacity and cost data
- Do not assume unlimited backup capacity
- Safety and quality requirements cannot be waived
- Document any unmet demand

Deliverables

- Dependency/process map
- Failure-impact calculations
- Three resilience concepts
- Decision matrix
- Response playbook
- Reflection

Reflection Questions

- 1 Which dependency was most critical?
- 2 Why is maximum inventory not always the best solution?
- 3 How did recovery time affect your choice?
- 4 What information should be monitored in real time?

Engineering Support

Engineering Tip

Resilience usually combines prevention, redundancy, flexible capacity, and a practiced response plan.

Common Mistake

A backup resource is not useful if it has never been qualified or cannot meet the required capacity.

Stretch Goal

Include a simultaneous cyber outage and supplier delay in the tabletop test.

Career Connection

Supply chain and business continuity engineers prepare factories for equipment, supplier, weather, and cyber disruptions.

Before You Submit

- Notebook evidence included
- Requirements checked
- Testing documented
- Reflection completed